Scalability& Future Plan

Date: 10 July 2026 **Team ID:** xxxxxxx **Project Name:** Credit Card Approval Prediction
Maximum Marks: 1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Single Flask dev-server process (no load balancing)	Limits concurrent request throughput	High
2	No authentication on the analyst dashboard	Anyone with the URL can submit predictions	Medium
3	Static model trained once, not retrained automatically	Model may drift as fraud patterns evolve over time	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local Flask process	Deploy behind Gunicorn/Nginx or via IBM Watson MLâ€™s managed scaling
2	Data Storage	No persistent DB; stateless scoring	Add a lightweight database for prediction/audit logs as usage grows
3	Performance	~45 req/sec on local dev server	Horizontal scaling via IBM Watson ML deployment or containerized replicas
4	Security	No auth layer	Add login-based access control and HTTPS enforcement

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add authentication & rolebased access for analysts	+1 month	Restrict access to authorized bank staff
Phase 3	Automated model retraining pipeline on new transaction data	+2 months	Keeps the model current as fraud patterns shift
Phase 4	Batch/CSV upload support and analyst review dashboard with logging	+3 months	Supports real bank-scale daily transaction review workflows